undervalue the quality of non-T5 articles relative to those published in the T5.[55] If such undervaluation exists, our analysis will understate the degree of comparability between T5 and non-T5 journals. Further, independent of quality, the T5 could attract more citations than field journals simply due to the fact that general interest journals are designed to target a wider audience than field journals. The reader should consider such potential biases when interpreting the results reported in this section.

4.1.1 Comparisons Against the Aggregate T5 Distribution

Figure 14 plots distributions of residual $ln(Citations + 1)$ for articles published between 2000–2010 in each of the thirty journals considered in our analyses.[56] The journal-specific distributions are overlaid on a shaded distribution that represents the distribution of residual citations for all articles published between 2000–2010 in the T5. The residualization adjusts log citations for exposure effects, and yields an exposure-adjusted measure that can be used to compare the performance of articles across publication cohorts.[57] The residuals are obtained by estimating an OLS regression of $ln(Citations + 1)$ on a third-degree polynomial for the number of years elapsed between the year of publication and 2018 (the year when citations

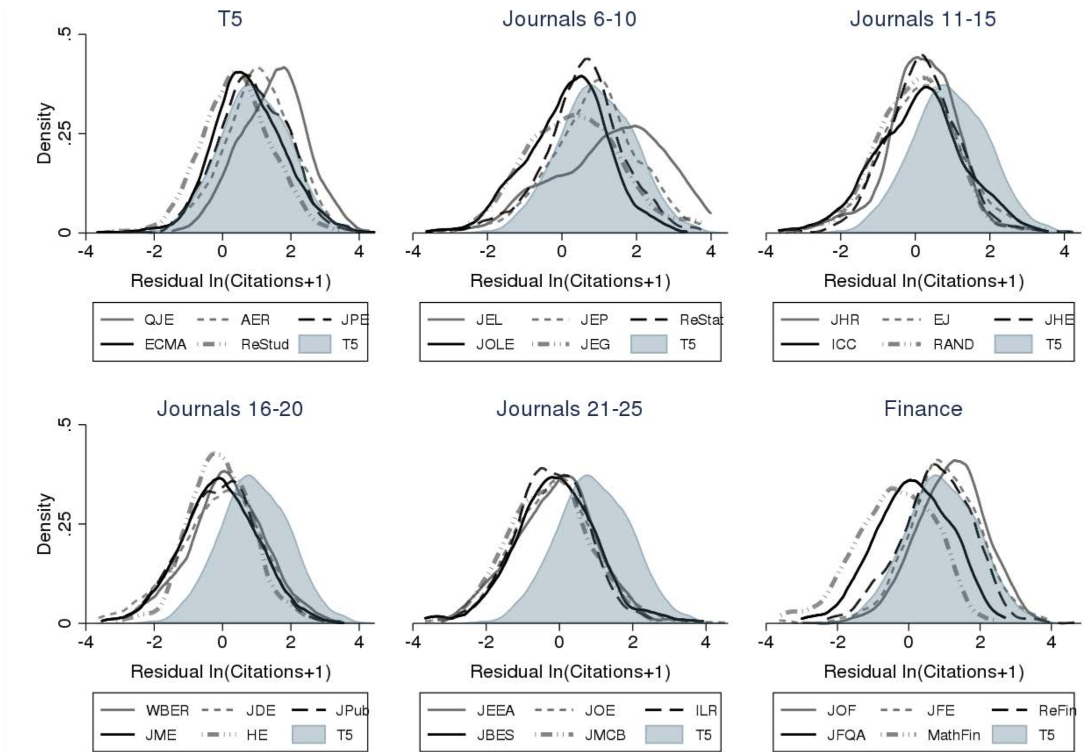
[55]Longstanding and deeply entrenched perceptions about the superiority of T5 publications serve to increase the visibility of T5 articles. In the presence of such differences, it is possible that T5 articles will attract more citations than non-T5 articles, conditional on article quality. The T5 journals are among the most popular and well-perceived journals in the profession. Analyzing the results of a survey of 92 Economists, Hawkins et al. (1973) show that the AER, ECMA, JPE, and QJE were the four most highly regarded journals in the late 1960's and early 1970's (ReStat was ranked 5th, and ReStud was ranked 6th). The perceived superiority of these four journals have persisted over time. Analyzing the results of 2,103 responses to an online survey sent to AEA members in 2002, Axarloglou and Theoharakis (2003) replicate the findings of Hawkins et al. (1973) and show that the AER, ECMA, JPE, and QJE have continued to be perceived as most influential in the early 2000s. To the extent that scholars prefer citing articles from journals that they perceive to be of the highest quality and influence, we should expect a negative bias against non-T5 citations. In other words, it is possible that holding constant both an article's quality and its relevance to the citing author's work, T5 articles receive more citations than non-T5 articles due to longstanding and deeply entrenched perceptions of the superiority of articles published in the T5.

[56]Similar to Hamermesh (2018), we exclude notes, comments, reports of editors, and papers published in the *AER*'s annual issue of *Papers & Proceedings*. We also exclude papers that are less than 10 pages in length.

[57]The present analysis focuses on comparisons of this year-adjusted measure. The interested reader is referred to Online Appendix Figures O-A25–O-A27 for analogous plots that are specific to articles published in 2000, 2005, and 2010 respectively.

were recorded).[58]

Figure 14: Distribution of Residual Log Citations for Articles Published between 2000–2010 (Measured Through July, 2018)



Source: Scopus.com; accessed 07/2018.
Definition of journal abbreviations: QJE–Quarterly Journal Of Economics, JPE–Journal Of Political Economy, ECMA–Econometrica, AER–American Economic Review, ReStud–Review Of Economic Studies, JEL–Journal Of Economic Literature, JEP–Journal Of Economic Perspectives, ReStat–Review Of Economics And Statistics, JEG–Journal Of Economic Growth, JOLE–Journal Of Labor Economics, JHR–Journal Of Human Resources, EJ–Economic Journal, JHE–Journal Of Health Economics, ICC–Industrial And Corporate Change, WBER–World Bank Economic Review, RAND–Rand Journal Of Economics, JDE–Journal Of Development Economics, JPub–Journal Of Public Economics, JOE–Journal Of Econometrics, HE–Health Economics, ILR–Industrial And Labor Relations Review, JEEA–Journal Of The European Economic Association, JME–Journal Of Monetary Economics, JRU–Journal Of Risk And Uncertainty, JInE–Journal Of Industrial Economics, JOF–Journal Of Finance, JFE–Journal Of Financial Economics, ReFin–Review Of Financial Studies, JFQA–Journal Of Financial And Quantitative Analysis, and MathFin–Mathematical Finance.

The subfigure labelled T5 reveals that the distribution of citations to *QJE* articles has a considerable rightward shift relative to the other T5 journals. A comparison of the median *QJE* residual against the distribution of residuals for all T5 publications reveals that the median-cited *QJE* article ranks at the 71^{st} percentile of all T5 publications in terms of

[58]Online Appendix Table O-A32 presents comparison of median residualized citations (aggregate T5 vs. individual journals) using residuals obtained from four different specifications. The first three columns present comparisons that use residuals obtained from an OLS of $ln(Citations) + 1$ on first-, second-, and third-degree polynomials of years of exposure respectively. The last column uses residuals obtained from estimating $ln(Citations)+1$ as a function of indicators for exposure. The results are robust to the alternative specifications.

residualized citations.[59] In terms of median citations, the *QJE* is followed by *AER*, *JPE*, *ECMA*, and *ReStud*, with the median-cited *ReStud* article reaching the 31^{st} percentile of T5 citations.

The subfigure labeled Journals 6–10 in Figure 14 plots distributions of residualized citations for the five non-T5 economics journals with the highest median citations. Among non-T5 economics journals, the greatest number of citations accrue to survey journals. Citations to *JEL* exhibit a considerable rightward shift relative to the T5 distribution. Online Appendix Table O-A31 shows that the median-cited *JEL* article ranks at the 70^{th} percentile of all T5 publications in terms of residual citations, which is one percentile below the ranking for the median-cited *QJE* article. *JEL* is followed by the *JEP*, whose median-cited article is ranked at the median of the T5 distribution. *ReStat* ranks first among the non-survey economics journals with its median citation reaching the 38^{th} percentile of all T5 citations. It outranks *ReStud* by 7 percentile points and underperforms *ECMA* by only 3 points. The list of the five highest-cited non-T5 economics journals is rounded out by *JEG* whose median-cited article is ranked at the 30^{th} percentile of T5 citations, *JOLE* which has an analagous ranking at the 25^{th} percentile, and *JHR, JHE*, and *ICC* which are all tied at the 24^{th} percentile.[60]

As previously noted, finance has emerged as an important subfield in economics. Not surprisingly, finance journals have lives of their own. They attract greater citations than non-T5, non-survey economics journals. *JOF* is most highly cited, with its median-cited article reaching the 61^{st} percentile of all T5 publications. It is followed by *JFE* and *ReFin*, both of which have median citations above that of *ECMA* and *ReStud*.

Finally, we note that the relative performance of non-T5 journals improves considerably when the comparison excludes the two highest-cited T5 journals (AER and QJE).

[59]See Online Appendix Table O-A31 for comparisons of journal-specific median citations against the T5 distribution of citations.

[60]The next three subfigures in Figure 14 present distributions for fifteen additional economics journals, listed in decreasing order of median citations. The first six of these journals have median-cited articles that are ranked at or above the 20^{th} percentile of all T5 articles.

Thus, while the median-cited *ReStat* article ranks in the 38^{th} percentile of the overall T5 distribution, its rank improves to the 48^{th} percentile when the comparison set is restricted to articles in *JPE*, *ECMA*, and *ReStud*. Similar improvements are recorded for other non-T5 journals. The reader is referred to Online Appendix Section 7.1 for further details on comparisons of non-T5 journals against subsets of the T5.

4.2 Which Journals Publish Influential Research Papers?

This section compares T5 and non-T5 journals with respect to the volume of influential articles published by each journal between the period 2000–2010. To proceed, we use the residualized citations used in the previous sections to group articles from the 30 economics journals into four performance-based bins: articles with the Top 25%, Top 10%, Top 5%, and Top 1% of residual citations. We then calculate the proportion of articles in each quality bin that was published by each of the 30 economics journals. Table 2 presents a ranking of the 30 journals based on unadjusted proportions.

The *AER* features prominently in these rankings, contributing the largest proportion of articles to each of the quality bins except the Top 1%. The *QJE* ranks second in the 25%, 10% and 5% bins, and ranks first in the Top 1% bin. With the exception of the Top 25% bin, *AER* and *QJE* account for a combined 30% or more of the articles in each citation bin (they account for 23.5% of the articles in the 25% bin). The other T5 journals contribute fewer influential articles.[61]

The non-T5 non-survey journals publish many influential articles. *JOE*, *ReStat* and *JEG* account for a combined 13.6% of all articles in the Top 1% of residual citations. The contributions from these three journals are not only significant in absolute terms, but also in relation to the T5. All three journals produce more Top 1% articles than *ReStud*, two of the

[61] With the exception of the 1% bin, *JPE* and *ECMA* are ranked within one point of each other. *ReStud*, on the other hand, ranks considerably lower than the other four T5 journals, and is outranked by many non-T5 journals in all four categories. The appearance of *ReStud* as an outlier among the T5 is consistent with the findings from the previous sections that show that the median *ReStud* article was ranked in the 31^{st} percentile of all T5 publications in terms of residual citations.

Table 2: Publication Volume-Unadjusted Proportion of Influential Articles Published By Individual Journals Between 2000–2010

Rank	**Top 25% Citations N=3321**		**Top 10% Citations N=1329**		**Top 5% Citations N=665**		**Top 1% Citations N=133**	
1.	**AER**	(14.0%)	**AER**	(16.6%)	**AER**	(17.7%)	**QJE**	(17.3%)
2.	**QJE**	(9.5%)	**QJE**	(14.0%)	**QJE**	(15.6%)	JEL	(13.5%)
3.	**ECMA**	(6.7%)	JEP	(7.6%)	JEP	(9.6%)	**AER**	(12.8%)
4.	JEP	(6.6%)	**ECMA**	(7.4%)	JEL	(8.0%)	JEP	(9.8%)
5.	**JPE**	(5.7%)	**JPE**	(6.8%)	**ECMA**	(7.1%)	**ECMA**	(8.3%)
6.	EJ	(5.2%)	JEL	(5.5%)	**JPE**	(5.1%)	JOE	(5.3%)
7.	JOE	(5.2%)	ReStat	(4.5%)	JOE	(4.7%)	ReStat	(4.5%)
					ReStat			
8.	ReStat	(4.8%)	EJ	(4.4%)		(.%)	JEG	(3.8%)
							JPE	
9.	JPub	(4.5%)	JOE	(4.2%)	EJ	(4.5%)		(.%)
10.	JME	(3.8%)	**ReStud**	(3.5%)	JME	(2.6%)	EJ	(3.0%)
					ReStud		JHE	
							RAND	
11.	JDE	(3.8%)	JPub	(3.1%)		(.%)		(.%)
12.	**ReStud**	(3.7%)	ICC	(3.0%)	ICC	(2.4%)		(.%)
					JPub			
13.	JHE	(3.3%)	JDE	(2.7%)		(.%)	JBES	(2.3%)
							JEEA	
							JPub	
14.	JEL	(3.3%)	JME	(2.5%)	JHE	(2.0%)		(.%)
15.	ICC	(2.7%)	JHE	(2.2%)	JBES	(1.4%)		(.%)
					JEG			
16.	HE	(2.5%)	HE	(2.0%)		(.%)	ICC	(1.5%)
							ReStud	
17.	JMCB	(2.4%)	JMCB	(1.5%)	HE	(1.2%)		(.%)
					JDE			
					JOLE			
					RAND			
18.	JHR	(2.1%)	JEG	(1.4%)		(.%)	JME	(0.8%)
			JOLE				JOLE	
							WBER	
19.	RAND	(2.0%)		(.%)		(.%)		(.%)
20.	JOLE	(1.9%)	JEEA	(1.3%)		(.%)		(.%)

Source: Scopus.com; accessed 07/2018
Note: This table presents publication volume-unadjusted proportions of highly cited articles published by different journals.
Definition of journal abbreviations: QJE–Quarterly Journal Of Economics, JPE–Journal Of Political Economy, ECMA–Econometrica, AER–American Economic Review, ReStud–Review Of Economic Studies, JEL–Journal Of Economic Literature, JEP–Journal Of Economic Perspectives, ReStat–Review Of Economics And Statistics, JEG–Journal Of Economic Growth, JOLE–Journal Of Labor Economics, JHR–Journal Of Human Resources, EJ–Economic Journal, JHE–Journal Of Health Economics, ICC–Industrial And Corporate Change, WBER–World Bank Economic Review, RAND–Rand Journal Of Economics, JDE–Journal Of Development Economics, JPub–Journal Of Public Economics, JOE–Journal Of Econometrics, HE–Health Economics, ILR–Industrial And Labor Relations Review, JEEA–Journal Of The European Economic Association, JME–Journal Of Monetary Economics, JRU–Journal Of Risk And Uncertainty, JInE–Journal Of Industrial Economics, JOF–Journal Of Finance, JFE–Journal Of Financial Economics, ReFin–Review Of Financial Studies, JFQA–Journal Of Financial And Quantitative Analysis, and MathFin–Mathematical Finance.